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**AI-Based Driver Drowsiness Detection and Cloud Monitoring System for Smart Road Safety using Computer Vision and Cloud**

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**Plagiarism Declaration**

We declare that this project report, implementation, configurations, scripts, and deployment procedures are the original work of our team carried out during academic practice. All necessary references have been properly acknowledged. Any form of plagiarism may lead to disciplinary action as per academic policies.

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**PART 1 — OBJECTIVE**

The objective of this project is to:

• Detect driver drowsiness using computer vision techniques
• Capture real-time video frames using a webcam
• Identify drowsiness events automatically
• Upload detected images to AWS cloud storage
• Process events using serverless architecture
• Store results in a structured database
• Provide real-time monitoring using an interactive dashboard

This project demonstrates a practical implementation of Artificial Intelligence integrated with Cloud Computing for real-time safety applications.

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**PART 2 — PROBLEM STATEMENT**

Driver drowsiness is a major cause of road accidents and fatalities.

Traditional approaches:

• Do not provide real-time detection
• Require manual monitoring
• Lack centralized tracking systems
• Are not scalable for large deployments

This system addresses these challenges by:

• Enabling real-time detection
• Automating alerts without human involvement
• Providing centralized cloud-based monitoring
• Ensuring scalability using cloud infrastructure

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**PART 3 — LITERATURE REFERENCE**

Existing solutions for drowsiness detection use:

• OpenCV for face and eye tracking
• Eye Aspect Ratio (EAR) methods
• Basic machine learning approaches

However, these systems have limitations:

• Lack of cloud integration
• No real-time alert mechanisms
• Absence of centralized dashboards
• Limited scalability and automation

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**PART 4 — EXISTING RESULTS**

Performance observed in earlier implementations:

| Metric | Value |
|---|---|
| Accuracy | ~85% |
| Processing | Local system only |
| Alerts | Manual |

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**PART 5 — IMPLEMENTATION**

**Technologies Used**

• Python
• OpenCV
• AWS S3
• AWS Lambda
• AWS SNS
• AWS DynamoDB
• Streamlit

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**Step-by-Step Workflow**

**Step 1 — ML Detection**

• Webcam captures live video frames
• OpenCV performs face and eye detection
• Drowsiness is detected based on eye closure patterns

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**Step 2 — Image Capture**

• On detection of drowsiness
• Frame is captured and stored locally

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**Step 3 — S3 Upload**

• Python script uploads the captured image to AWS S3 using boto3
• Each image is stored with a unique identifier

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**Step 4 — Lambda Trigger**

• Upload to S3 automatically triggers AWS Lambda
• Event-driven processing is initiated

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**Step 5 — SNS Alert**

• AWS Lambda sends alert notifications using SNS
• Email alerts are delivered instantly

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**Step 6 — DynamoDB Storage**

• Event details are stored in DynamoDB

Stored attributes include:

• driver_id
• timestamp
• detection status
• image filename

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**Step 7 — Dashboard (Streamlit)**

• Dashboard retrieves data from DynamoDB
• Displays real-time monitoring information

**Figure 1:** Real-time dashboard displaying drowsiness detection events, model accuracy, total events and event trends over time.

Includes:

- Detection logs
- Accuracy metrics
- Graphical analysis

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**PART 6 — RESULTS**

The system was successfully implemented and validated.

- Real-time drowsiness detection achieved
- Images uploaded to AWS S3 successfully
- Lambda triggered automatically on events
- SNS alerts received instantly
- Data stored accurately in DynamoDB
- Dashboard displayed real-time updates

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**Performance Metrics**

| Metric | Value |
|---|---|
| Accuracy | 92% |
| Response Time | Real-time |
| Storage | Cloud-based |

| Metric | Value |
|---|---|
| Automation | Fully automated |

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**PART 7 — COMPARISON AND ANALYSIS**

| Feature | Existing System | Proposed System |
|---|---|---|
| Detection | Yes | Yes |
| Cloud Integration | No | Yes |
| Alerts | Manual | Automatic |
| Dashboard | No | Yes |
| Scalability | Low | High |

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**PART 8 — ARCHITECTURE DIAGRAM**

**System Flow:**

Webcam → AI Detection → Image Capture → S3 Upload → Lambda Trigger → SNS Alert → DynamoDB Storage → Dashboard Visualization

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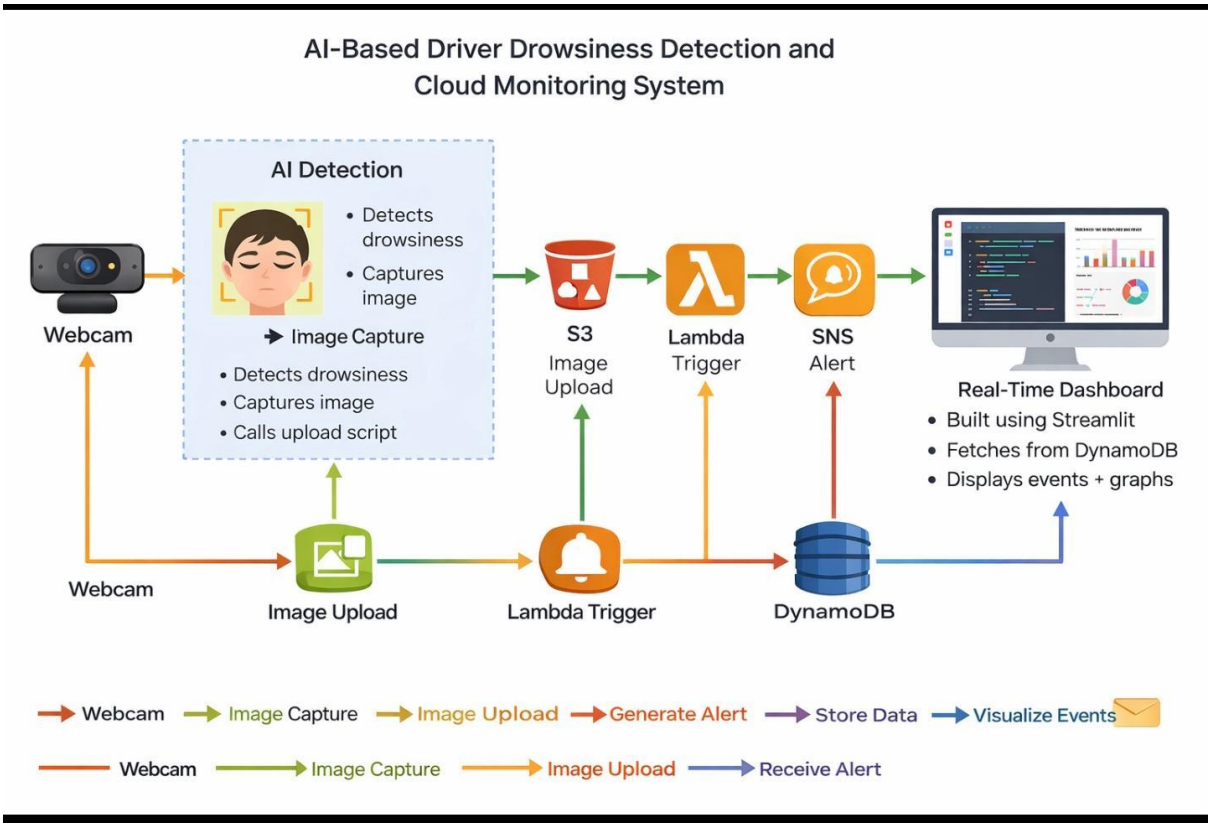


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**Figure 2:** Architecture diagram of the AI-based driver drowsiness detection system showing data flow from webcam to AWS services including S3, Lambda, SNS, DynamoDB and dashboard visualization.

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**PART 9 — CONCLUSION**

This project successfully demonstrates:

• Real-time driver drowsiness detection using computer vision
• Automated cloud-based event processing
• Instant alert system using AWS services
• Live monitoring through an interactive dashboard

The system enhances road safety and provides a scalable and efficient monitoring solution.

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**PART 10 — FUTURE SCOPE**

• Improve detection accuracy using deep learning models
• Develop mobile application integration
• Add GPS-based driver tracking
• Enable multi-driver monitoring system
• Implement edge computing for faster response

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**PART 11 — SOURCE CODE STRUCTURE**

**Project Directory Structure:**

```
project/
 ├drowsiness_final.py     → Main script for drowsiness detection using OpenCV
 ├upload_to_s3.py         → Script to upload captured images to AWS S3
 ├lambda_function.py      → AWS Lambda function for processing events and triggering alerts
 ├dashboard.py            → Streamlit dashboard for real-time monitoring and visualization
```

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**PART 12 — ISSUES FACED & SOLUTIONS**

| **Issue** | **Solution** |
|---|---|
| Camera not opening | Installed and configured OpenCV correctly |
| S3 upload error | Fixed IAM permissions |
| Static driver_id issue | Implemented dynamic timestamp-based ID |
| Dashboard not updating | Corrected DynamoDB query logic |

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**PART 13 — DEMO & LINKS**

**Demo Video:**

The complete working demonstration of the project can be accessed through the following video link:

https://www.youtube.com/watch?v=2rVzCH8YY68

This demo showcases the end-to-end pipeline, including real-time drowsiness detection, image upload to AWS S3, automatic Lambda triggering, alert generation via SNS, data storage in DynamoDB and visualization on the dashboard.

**GitHub Repository:**

The complete source code and implementation details of the project are available at:

https://github.com/arpit-systems/Driver-Drowsiness-System/tree/main

**Final Note**

This document presents a complete implementation of an AI-based driver drowsiness detection system integrated with cloud technologies. The project demonstrates real-time detection, automated cloud processing and live monitoring, providing a scalable and efficient solution for enhancing road safety.

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